

Full-Automation Testing — HelixVPN nano-detail spec (Volume 8 · §11.4.169 type 4)

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Rev 3 (2026-07-04): independently re-verified against SPECIFICATION.md / decision-register.md / 10-testing-acceptance-and-qa.md during a corpus-wide gap-analysis pass; the 9-AC DoD count is consistent with the overview and no new contradictions were found. Revision bumped for the audit pass itself.

Reconciled (§11.4.35, 2026-06-26): the MVP Definition-of-Done bank has **NINE** acceptance criteria (AC1-AC9) per the canonical overview 10-testing-acceptance-and-qa.md §7.2. The earlier "8-AC DoD bank" wording (§2, §6) was a miscount; an FA gate driving only 8/9 ACs is a §11.4.118 coverage bluff. Corrected to the 9-AC bank.

Nano-detail expansion of §5.4 of the Volume-8 overview. Full-automation (FA) is the §11.4.98 invariant made concrete: **every** E2E / integration / Challenge must be re-runnable end-to-end with **zero human intervention** after startup, N=3 consecutively, with self-cleaning state and an identical PASS each time (§11.4.50). For HelixVPN the whole MVP-DoD demo — helixvpnctl init → enroll connector+client → reach → deny → escalate → revoke → prove no-leak — runs as **one command**. The single permitted human touch-point is credential bootstrap *outside* execution (a .env token, §11.4.10). This document fixes the FA harness, the determinism wrapper, the self-driving requirement, the captured evidence, the gate, the paired

§1.1 mutation, and skeletons. Spec-only; unproven assumptions marked UNVERIFIED. Siblings: [unit.md](#), [integration.md](#), [e2e.md](#), [challenges.md](#), [helixqa.md](#).

Table of contents

- [1. Scope — what FA covers \(the §11.4.98 invariant\)](#)
 - [2. Harness — make spike / make qa one-shot drivers](#)
 - [3. Fixtures — real system end-to-end, no human in the loop](#)
 - [4. Evidence taxonomy — what an FA PASS captures](#)
 - [5. Determinism — the `ab\_run\_n` times contract](#)
 - [6. Acceptance gate — when FA blocks a release](#)
 - [7. The paired §1.1 mutation \(anti-bluff proof\)](#)
 - [8. Test skeletons](#)
 - [9. Open decisions surfaced for QA](#)
 - [Sources verified](#)
-

1. Scope — what FA covers (the §11.4.98 invariant)

FA is not a *new* set of assertions — it is a **property** every higher test type must satisfy: self-driving end-to-end, re-runnable forever, no human action after startup. The §11.4.98 forensic anchor: *"these tests MUST be able to rerun endless times ... no false positives ... every test obtains real proofs of everything working."*

FA-covered flow	Composed from	The §11.4.98 property proven
Full MVP-DoD demo	init → enroll → reach (AC2) → deny (AC3) → escalate (AC4) → reconcile (AC5) → revoke (AC6) → kill-switch (AC7)	the whole product self-hosts + works with one command, N=3
Self-host from zero (AC1)	helixvpctl init + stack start, healthchecks green	a fresh operator reproduces the deploy autonomously
Every E2E (e2e.md)	the netns rig flows	each E2E re-runs with no human typing
Every INT (integration.md)	containers-booted seams	infra boots itself; no manual podman
	the DoD bank	the challenges engine drives +

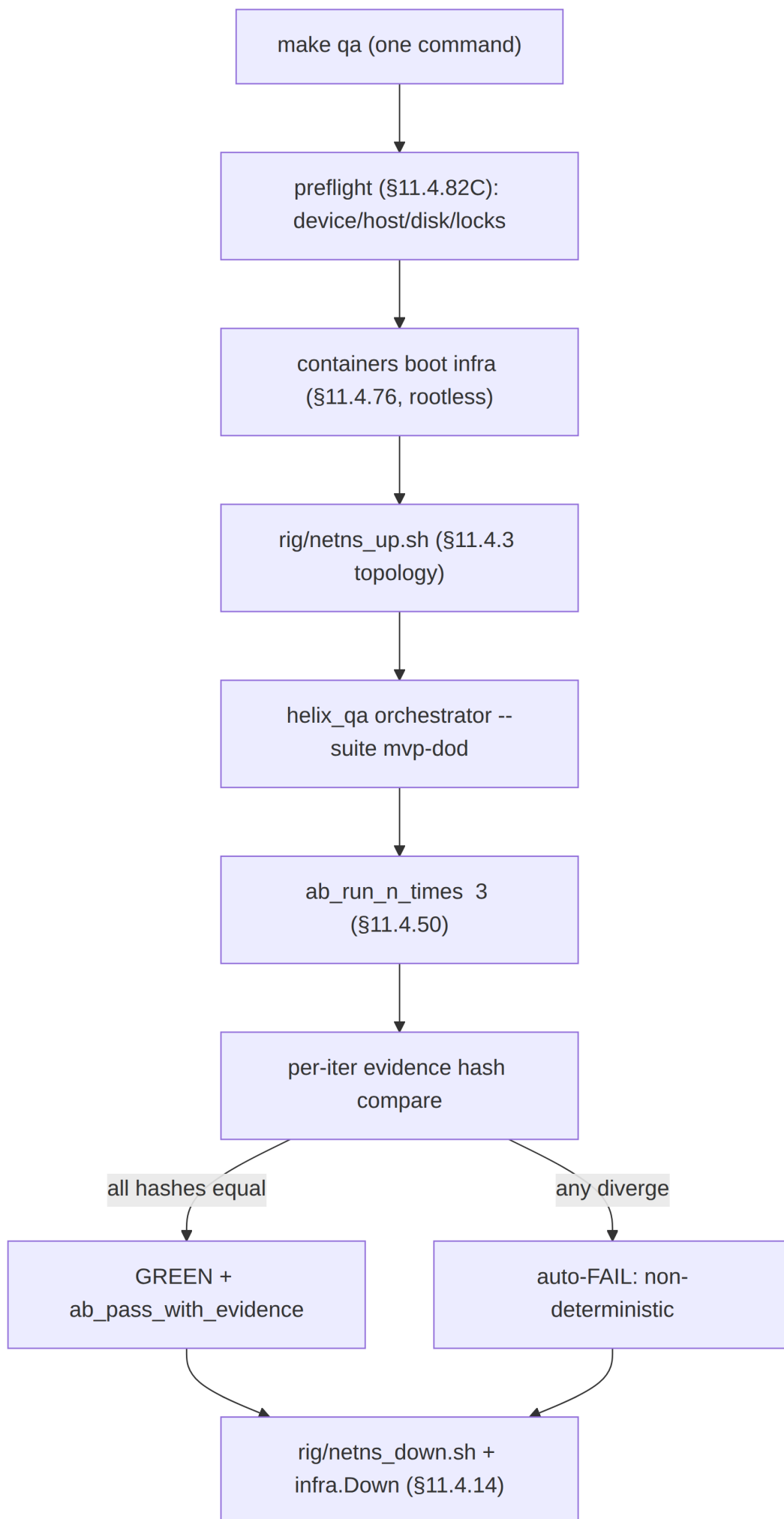
FA-covered flow	Composed from	The §11.4.98 property proven
Every Challenge (challenges.md)		scores autonomously
FFI round-trip (G5)	Dart calls <code>connect()</code> , receives the status stream	a UI-less driver exercises the FFI [04_P0 G5]

The forbidden human touch-points (any of these makes a test a §11.4.98 PASS-bluff at the automation layer): an operator typing a value mid-run; a manual UI click a test depends on; a hand-triggered webhook; a 60 s "wait for the human to respond" window (also a §11.4.50 determinism violation); a hard-coded session UUID that collides with a live dev session ([the Herald §11.4.98(C) lesson]). The **only** permitted out-of-band step is one-time credential bootstrap (a `.env` token, an SSH key) — configuration, not test driving.

2. Harness — make spike / make qa one-shot drivers

Two one-shot Make targets are the FA drivers ([overview §9](#)):

Target	Phase	What it drives
make spike	Phase 0	the G1–G6 gate sweep (rig reach, MASQUE-through-DPI, FFI round-trip, push-reconcile) end-to-end, N=3
make qa	Phase 1	the <code>helix_qa</code> autonomous session over the 9-AC DoD bank (helixqa.md)



The driver chains: a **preflight** (§11.4.82(C) — device/sink reachability, host memory/disk, no stale locks, no orphan processes — a 30 s check that prevents a 45-minute run against a broken precondition); **infra boot** via containers (§11.4.76, rootless §11.4.161); **rig up** (§11.4.3 topology); the **orchestrated suite** under the determinism wrapper; **teardown** on every exit path (§11.4.14). Long suites run **backgrounded** (nohup ... & disown, §11.4.89) so the main work stream is never blocked while FA runs.

3. Fixtures — real system end-to-end, no human in the loop

Fixture	Real or mock	Notes
control plane + edge + client	real	the full stack, booted by the driver
infra (PG/Redis)	real , via containers	on-demand-infra invariant (integration.md §2)
netns rig	real	the E2E substrate (e2e.md §2)
.env credential bootstrap	real token, out-of-band	§11.4.10 — the one permitted human step, <i>before</i> execution
programmatic drivers	real	helixvpncctl, curl, helix-core CLI — never a human keystroke
mocks / manual steps	FORBIDDEN (§11.4.27 / §11.4.98)	a manual click or a mock makes the suite non-re-runnable

FA explicitly **drives** the UI programmatically where a UI flow is in scope — through panoptic ([overview §5.14](#)) or, for non-introspectable surfaces, the §11.4.117 CV/OCR pixel oracle — never "the operator clicks Connect". A flow that genuinely cannot be driven autonomously (a hard human-only login, a CAPTCHA) is an honest operator_attended SKIP with a tracked migration item (§11.4.52), **never** a faked PASS.

4. Evidence taxonomy — what an FA PASS captures

FA's distinguishing evidence is the **re-runnability proof** on top of each composed test's own §11.4.69 evidence:

FA evidence	Produced by	Proves
N=3 identical evidence-hash log	ab_run_n_times	§11.4.50 determinism (no flake escape)
one-command transcript	the driver	§11.4.98 self-driving (no human step in the log)
self-cleaning proof	post-run sanity check	the host returned to quiescence (§11.4.14)
per-AC §11.4.69 artifacts	the composed E2E/INT/CHAL	the underlying features actually work
helix_up==1 healthcheck	AC1 driver	the stack self-hosted from zero

Per §11.4.69 every FA PASS flows through `ab_pass_with_evidence` citing the N=3 hash log + the composed artifacts; the **absence of any human-action line in the run transcript** is itself part of the evidence (a §11.4.98 `grep` for "operator MUST" / manual-prompt patterns in the driver is a pre-build lint).

5. Determinism — the `ab_run_n_times` contract

The §11.4.50 wrapper is the heart of FA: it loops the suite, captures an evidence-hash per iteration, and FAILs on **any** divergence — there is no "first-pass-was-a-flake" path.

```
# the §11.4.50 determinism wrapper every FA suite uses ([overview §5.4])
ab_run_n_times() {
    local name="$1" n="$2" fn="$3" h prev=""
    for i in $(seq 1 "$n"); do
        "$fn" "$i"
        h=$(sha256sum "qa-results/$name/run_$i/evidence.json" | cut
            -d' ' -f1)
        [ -z "$prev" ] || [ "$h" = "$prev" ] || ab_fail "non-
            deterministic: run $i hash diverged from run $((i-1))"
        prev="$h"
    done
    ab_pass_with_evidence "$name x$n deterministic" "qa-results/
        $name/"
}
```

- **N policy:** N=3 for normal FA runs, **N=10** for cycle-validation suites (the pre-tag §11.4.40 sweep). The same artifact MD5 + same rig + same topology each iteration.
- **What hashes:** the *evidence content* (pcap structural hash, rowset, body hash, delta JSON) — **not** wall-clock timing (which is captured as a value compared to an SLO budget, not

asserted equal). A timing field is normalised out of the hashed evidence.json before hashing.

- **Self-cleaning state** is the precondition for determinism: each iteration starts from netns_up.sh + a fresh containers boot, so run $i+1$ cannot inherit run i 's tunnel/route/row (§11.4.108/.139 clean-artifact, defeats B4 stale-state).
- A 9/10 PASS is **auto-FAIL** — a real defect (a hidden timing/ordering dependency), never a flake (§11.4.50 mechanises §11.4.7 demotion-evidence).

6. Acceptance gate — when FA blocks a release

Gate	Bar	Bound item
make spike	G1-G6 all clear N=3, evidence captured	Phase-0 exit (overview §7.1)
make qa	the 9-AC DoD bank PASS via helix_qa, N=3	Phase-1 DoD (overview §7.2)
CM-FA-RERUNNABLE (§11.4.98)	every E2E/INT/CHAL passes at -count=3; no human-action line in any driver	pre-build + runtime
CM-FA-NO-MANUAL-STEP	no operator MUST / manual-prompt / hard-coded-session-UUID pattern in test sources	pre-build lint

FA is the **operating-mode gate**: a feature whose only validation path is operator-attended is a §11.4.52 release blocker (its coverage-ledger cell sits at OPERATOR_ATTENDED_ONLY until promoted with a tracked migration item, [overview §6](#)). The §11.4.40 full-suite-retest-before-tag **is** an FA run at N=10 — a release tag is created only after the whole MVP-DoD demo runs autonomously, deterministically, on a clean baseline (§11.4.126 release-scope terminal condition).

7. The paired §1.1 mutation (anti-bluff proof)

```
# §1.1 mutation for CM-FA-RERUNNABLE (proves the determinism
wrapper actually catches drift)
- inject a non-deterministic byte into evidence.json (e.g. an un-
normalised timestamp
  or a random run-id) BEFORE hashing
```

- assert: `ab_run_n_times` reports "non-deterministic: run 2 hash diverged" → FA FAILs
- restore the normalisation; assert N=3 identical → PASS

§11.1 mutation for CM-FA-NO-MANUAL-STEP (proves the no-human-step lint is real)

- add a line ``read -p "operator: type the token> " TOK`` into a driver
- assert: the pre-build lint flags the manual-prompt pattern → gate FAILs
- restore (token from `.env §11.4.10`); assert PASS

These mutations prove the FA gates are not tautologies: the determinism wrapper genuinely FAILs on injected drift, and the no-manual-step lint genuinely catches a human touch-point. The mutation/restore cycle is serialised + working-tree-verified clean before any unrelated commit (§11.4.84).

8. Test skeletons

8.1 The whole MVP-DoD demo as one self-driving function

```
# scripts/fa_mvp_dod.sh – driven by `make qa`; ZERO human steps
                        after startup (§11.4.98)
fa_mvp_dod() {
    iteration index
    local run="qa-results/fa_mvp_dod/run_$1"; mkdir -p "$run"
    preflight_or_fail
    §11.4.82C: host/disk/locks/reachability
    trap 'rig/netns_down.sh; infra_down' EXIT
    # §11.4.14 quiescence on every exit

    # AC1 – self-host from zero (token from .env, §11.4.10 –
    # the ONE out-of-band step)
    helixvpncctl init --config "$run/helix.yaml" --token-from-env
    wait_healthy helix_up 60 || ab_fail "AC1: stack not healthy"

    # AC2 enroll + reach, AC3 deny, AC4 escalate, AC5
    # reconcile, AC6 revoke, AC7 kill-switch
    rig/enroll.sh connector client | tee "$run/enroll.log"
    rig/test_reach.sh pass ; cp qa-results/e2e/
    *reach.pcap "$run/" # AC2
    rig/test_reach.sh
    deny
    # AC3
    rig/escalate.sh
    AC4
```



```

rig/reconcile.sh
    AC5
rig/revoke.sh
    AC6
rig/killswitch_drop.sh
    AC7 (SEC §5.7)

# single normalised evidence manifest the determinism wrapper
    hashes
collect_evidence "$run" > "$run/evidence.json"          # timing
    fields normalised out
}

# the one command: make qa -> ab_run_n_times "fa_mvp_dod" 3
    fa_mvp_dod

```

8.2 AC1 self-host healthcheck poll (loading is a distinct state)

```

# wait_healthy – poll a real healthcheck, never a fixed sleep
    (§11.4.107(3))
wait_healthy() {                                         # $1
    metric $2 timeout_s
    local t0; t0=$(date +%s)
    until [ "$(curl -s localhost:9090/metrics | grep "^$1 " | awk
        '{print $2}')" = "1" ] ; do
        sleep 0.5
        [ $(( $(date +%s) - t0 )) -lt "$2" ] || return 1    #
        timeout+unreachable handled by caller (SKIP vs FAIL)
    done
}

```

8.3 FFI round-trip (G5) — Dart drives the core, no UI

```

// helix-ui/integration_test/ffi_roundtrip_fa_test.dart (G5,
    §11.4.98 self-driving)
// Drives the Rust core through flutter_rust_bridge with NO
    widget, NO human tap.
void main() {
    test('FA: connect() returns a live status stream, N-runnable',
        () async {
            final core = HelixCore.bridge();                // real
            FFI, real core
            final statuses = <TunnelStatus>[];
            final sub = core.statusStream().listen(statuses.add);
            await core.connect(ConnectArgs(target:
                '10.10.0.20')); // real connect, real tunnel
            await pumpUntil(() =>
                statuses.contains(TunnelStatus.connected),
                timeout: const Duration(seconds: 8)); //
            loading distinct (§11.4.107(3))
        }
    );
}

```

```

    expect(statuses, contains(TunnelStatus.connecting)); // saw
        the loading state
    expect(statuses.last, TunnelStatus.connected);
    await core.disconnect(); await sub.cancel(); // self-
        cleaning (§11.4.14)
    writeEvidence('qa-results/fa/ffi_roundtrip', statuses);
  });
}

```

8a. Self-cleaning state — the re-runability precondition

FA's "re-runnable forever" property is impossible without **self-cleaning state**: run $i+1$ must not inherit anything from run i . The catalog of state every FA flow must reset, and how:

State class	Left by a run	Reset mechanism	Constitution
network namespaces / veths / routes	the netns rig	trap 'rig/netns_down.sh' EXIT	§11.4.14
nft DPI table + tc netem qdisc	dpi_block.sh / impair.sh	netns_down.sh flushes both	§11.4.14
containers (PG/Redis/edge)	containers/pkg/boot	infra.Down(ctx) on every exit path	§11.4.14 / §11.4.76
overlay tunnel + WG peers	helix-core	helix-core disconnect + a clean re-deploy	§11.4.108/.139
DB rows (tenants/devices/allocations)	the seed step	fresh containers boot each iteration	§11.4.50
qa-results/raw recordings (the agent's own, in-scope)	panoptic capture	fresh-corpus rotation before a new run	§11.4.154(B)
stale locks / orphan processes	a crashed prior run	preflight scan (§11.4.82(C)/(H))	§11.4.82

The post-run **sanity check** asserts the host returned to quiescence; a surviving namespace, container, qdisc, or orphan process FAILs the just-completed FA run (§11.4.14) — a test that "passes" but leaves orphan state is itself a defect, because the

next run inherits it and the determinism contract silently breaks. Fresh-corpus rotation removes **only the agent's own in-scope** prior recordings (§11.4.154(B)); foreign/operator files are surfaced, never deleted (§11.4.122 / §9.2).

8b. The §11.4.98 compliance audit (existing-test classification)

§11.4.98(C)(5) requires every test be classified COMPLIANT vs NON-COMPLIANT against the no-manual-step invariant, and a NON-COMPLIANT test rewritten within the grace window or graduated to §11.4.90 obsolete citing §11.4.98. The HelixVPN audit lens:

Pattern (NON-COMPLIANT)	HelixVPN remediation
"operator types the enrollment token"	token from .env (§11.4.10), helixvpnctl enroll-token programmatic
"operator taps Connect in the app"	panoptic drives the tap; §11.4.117 pixel oracle if the UI tree is blank
"operator confirms the shield is green"	vision_engine OCR verdict + core-FSM cross-check (helixqa.md)
"operator triggers the DPI block"	rig/dpi_block.sh applies the nft rule autonomously
hard-coded session UUID colliding with a live session	a per-run unique id minted at startup (the Herald §11.4.98(C) lesson)
60 s "wait for human response" window	replaced by a real healthcheck poll (§8.2), bounded by a timeout

9. Open decisions surfaced for QA

#	Decision	Options	Recommendation
FA-D1	N for the pre-tag sweep	N=3 vs N=10	N=10 for the §11.4.40 pre-tag cycle-validation , N=3 for routine FA — the cost is amortised against a release, not every push
FA-D2	UI-flow driver in FA	panoptic drive vs CV/OCR pixel oracle vs operator-attended	panoptic where the Flutter hierarchy is introspectable; §11.4.117 pixel oracle for canvas/leanback; operator-attended SKIP only when

#	Decision	Options	Recommendation
			genuinely infeasible (§11.4.52)
FA-D3	timing-field hashing	include timing (over-strict) vs normalise out	normalise timing out of evidence.json before hashing; compare timing separately to the SLO budget — avoids false non-determinism FAILs (§11.4.50)

9a. FA vs the other types — the composing relationship

FA is a **property layered onto** the other types, not a parallel suite:

Type	What FA adds to it
UNIT	ab_run_n_times determinism even at the unit layer (seeded property tests)
INT	the on-demand-infra boot is part of the one command — no manual podman
E2E	each rig flow re-runs N=3 with self-cleaning netns + identical pcap verdict
CHAL	the challenges engine runs unattended; no human triggers a driver
HQA	the whole autonomous session is the FA driver at the orchestration layer

The §11.4.98 invariant is therefore tested *transitively*: CM-FA-RERUNNABLE re-runs the entire pyramid at -count=3 and asserts every layer is self-driving; a single manual touch-point anywhere fails the whole FA gate, regardless of how green the individual layer looked. This is why FA is a §11.4.52 release blocker for any feature whose only path is operator-attended — the coverage-ledger cell cannot reach AUTONOMOUS_VERIFIED until the FA property holds end-to-end.

UNVERIFIED: the `helixvpctl init --token-from-env flag name` and the `localhost:9090/metrics helix_up metric name` are taken from the overview/[06] skeleton; confirm against [../v06-deploy/helixvpctl.md](#) and [../v03-control-plane/svc-telemetry.md](#) at implementation time.

Sources verified

- Volume-8 overview §2 (taxonomy row FA), §5.4 (ab_run_n_times wrapper), §5.6 (helix_qa make qa), §7.1 (G5), §7.2 (AC1-AC9), §9 (Makefile qa:) — read 2026-06-26.
- Sibling specs cross-referenced: e2e.md, integration.md, challenges.md, helixqa.md, ../v06-deploy/helixvpnctl.md, ../v03-control-plane/svc-telemetry.md, ../v04-client/ffi-surface.md — filenames confirmed present (v06/v03 [GEN]-planned) 2026-06-26.
- Constitution: §11.4.98 (full-automation / re-runnable / no manual step / no false positive), §11.4.50 (determinism / ab_run_n_times), §11.4.52 (autonomous-validation / operator-attended is a blocker), §11.4.117 (CV/OCR pixel oracle), §11.4.10 (credential bootstrap out-of-band), §11.4.14 (cleanup quiescence), §11.4.82(C) (preflight), §11.4.89 (background execution), §11.4.40 (pre-tag full-suite retest), §11.4.108/139 (clean-artifact), §11.4.126 (release-scope terminal condition), §1.1 (paired mutation) — from CLAUDE.md in-context.
- The Herald §11.4.98(C) hard-coded-session-UUID lesson is cited from the constitution text in-context — not a HelixVPN artifact.